

Technical Project Report: Detection and Analysis of a Windows Host Malware Infection Involving Remcos RAT, Cridex, and Quasar RAT, Using IDS Alert Correlation, File Hash Analysis, and Threat Intelligence Lookup in a Virtualized SOC Lab Environment

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1. Introduction

In this project, I investigated a Windows host malware infection using Security Onion, simulating a real SOC analyst's workflow from initial alert discovery through full attack timeline reconstruction. The scenario involved a Windows computer that triggered automated security alerts on 19 March 2019, indicating a possible malware infection. Using Sguil to review the raw IDS alerts and Kibana to correlate the broader event timeline, I worked to identify the exact time of the attack, the infected host and user, and the specific malware responsible for the compromise.

2. Virtual Lab Environment Setup

Tools and Software

- VirtualBox (Oracle)
- CyberOps Security Onion
- Ubuntu (64-bit) Operating System
- Sguil
- Kibana
- Wireshark

Part 1: Investigating the Attack with Sguil and Wireshark

Starting in Sguil, I located the group of alerts from 19 March 2019 to begin tracing the sequence of events tied to this attack.

ApplicationsPlacesSguil.tkThu 15:05

SGUIL-0.9.0 - Connected To localhost

FileQueryReportsSound: OffServerName: localhostUserName: analystUserID: 22026-06-25 15:05:29 GMT

RealTime EventsEscalated Events

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	66	seconl...	5.165	2018-08-11 05:30:06	192.168.1.95	50185	185.68.93.18	80	6	ET INFO GENERIC SUSPICIOUS POST to Dotted Qua...
RT	1	seconl...	5.439	2019-03-19 01:45:03	10.0.90.215	52609	10.0.90.9	53	17	ET POLICY DNS Update From External net
RT	2	seconl...	5.440	2019-03-19 01:47:04	10.0.90.215	49204	209.141.34.8	80	6	ET INFO Executable Download from dotted-quad Host
RT	1	seconl...	5.441	2019-03-19 01:47:04	10.0.90.215	49204	209.141.34.8	80	6	ET CURRENT_EVENTS Possible Malicious Macro DL E...
RT	2	seconl...	5.442	2019-03-19 01:47:04	209.141.34.8	80	10.0.90.215	49204	6	ET CURRENT_EVENTS Likely Evil EXE download from ...
RT	12	seconl...	5.444	2019-03-19 01:47:04	209.141.34.8	80	10.0.90.215	49204	6	ET CURRENT_EVENTS Likely Evil EXE download from ...
RT	12	seconl...	5.456	2019-03-19 01:47:04	209.141.34.8	80	10.0.90.215	49204	6	ET INFO SUSPICIOUS Dotted Quad Host MZ Response
RT	12	seconl...	5.468	2019-03-19 01:47:04	209.141.34.8	80	10.0.90.215	49204	6	ET POLICY PE EXE or DLL Windows file download HTTP
RT	404	seconl...	5.480	2019-03-19 01:49:45	10.0.90.215	49205	103.1.184.108	2404	6	ET TROJAN Remcos RAT Checkin 23
RT	1	seconl...	5.482	2019-03-19 01:49:46	10.0.90.215	49206	217.23.14.81	80	6	ET CURRENT_EVENTS Terse alphanumeric executabl...
RT	2	seconl...	5.483	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET CURRENT_EVENTS Likely Evil EXE download from ...
RT	12	seconl...	5.485	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET INFO EXE - Served Attached HTTP
RT	12	seconl...	5.497	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET INFO Packed Executable Download
RT	12	seconl...	5.509	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET CURRENT_EVENTS DRIVEBY Likely Evil EXE with ...
RT	12	seconl...	5.521	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET CURRENT_EVENTS Likely Evil EXE download from ...
RT	12	seconl...	5.533	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET POLICY Terse Named Filename EXE Download - P...
RT	16	seconl...	5.571	2019-03-19 02:03:24	31.22.4.176	3389	10.0.90.215	49213	6	ET TROJAN ABUSE.CH SSL Blacklist Malicious SSL cer...
RT	13	seconl...	5.589	2019-03-19 02:08:17	203.45.1.75	443	10.0.90.215	49218	6	ET TROJAN ABUSE.CH SSL Blacklist Malicious SSL cer...
RT	3	seconl...	5.942	2019-03-19 04:54:34	115.112.43.81	443	10.0.90.215	49289	6	ET TROJAN ABUSE.CH SSL Blacklist Malicious SSL cer...
RT	3	seconl...	5.1108	2019-04-15 16:42:14	10.0.90.175	55465	10.0.90.9	53	17	ET POLICY DNS Update From External net

IP ResolutionAgent StatusSnort StatisticsSystem MsgsUser Msgs

☐ Reverse DNS☒ Enable External DNS

Show Packet DataShow Rule

IPSource IPDest IPVerHLTOSlenIDFlagsOffsetTTLCkSum

Search Packet PayloadHexTextNoCase

SGUIL-0.9.0 - Connected To localhost1 / 4

First alert: 01:45:03. Last alert: 04:54:34. Most alerts cluster within a 20-minute window (01:47:04 to 02:08:17), with several sharing the exact same timestamp. This pattern suggests automated, scripted activity rather than manual behavior, consistent with a malware infection chain executing through multiple stages in rapid succession.

1.1 Initial Alert Review: Packet Header and Snort Rule Inspection (Alert ID 5.439)

The screenshot shows the SGUIL-0.9.0 interface. The top bar indicates 'Fri 04:21' and 'Connected To localhost'. The main pane displays a table of alerts. Alert ID 5.439 is highlighted, showing a message: 'ET POLICY DNS Update From ...'. The sidebar on the left contains filters for 'IP Resolution', 'Agent Status', 'Snort Statistics', and 'System Msgs'. The main pane also shows the triggered rule and packet data, including the source IP (10.0.90.215) and destination IP (10.0.90.9).

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	7	seconion-l...	5.431	2017-06-27 13:44:04	62.210.140.158	80	192.168.1.96	49250	6	ET TROJAN Pushdo.S CnC res...
RT	1	seconion-l...	5.438	2017-06-27 13:44:32	208.83.223.34	80	192.168.1.96	49932	6	ET POLICY TLS possible TOR ...
RT	3	seconion-l...	5.149	2018-08-11 05:15:17	192.168.1.95	54515	192.168.1.6	53	17	ET POLICY DNS Update From ...
RT	5	seconion-l...	5.150	2018-08-11 05:20:59	149.129.222.112	80	192.168.1.95	49335	6	ET INFO Packed Executable Do...
RT	5	seconion-l...	5.155	2018-08-11 05:20:59	149.129.222.112	80	192.168.1.95	49335	6	ET POLICY PE EXE or DLL Wl...
RT	5	seconion-l...	5.160	2018-08-11 05:20:59	149.129.222.112	80	192.168.1.95	49335	6	ET INFO Executable Retrieved ...
RT	66	seconion-l...	5.165	2018-08-11 05:30:06	192.168.1.95	50185	185.68.93.18	80	6	ET INFO GENERIC SUSPICIO...
RT	1	seconion-l...	5.439	2019-03-19 01:45:03	10.0.90.215	52609	10.0.90.9	53	17	ET POLICY DNS Update From ...
RT	2	seconion-l...	5.440	2019-03-19 01:47:04	10.0.90.215	49204	209.141.34.8	80	6	ET INFO Executable Download...
RT	1	seconion-l...	5.441	2019-03-19 01:47:04	10.0.90.215	49204	209.141.34.8	80	6	ET CURRENT_EVENTS Possi...

Selecting Alert ID 5.439 and enabling Show Packet Data and Show Rule revealed the triggered Snort signature: 'ET POLICY DNS Update from External net' (SID 2009702), flagging a DNS update sent from an external source to an internal host over UDP port 53. The packet header confirms the alert relates to DNS-based activity, but the raw payload alone isn't enough to determine intent, so I pivoted to Wireshark for a closer look at the actual communication.

1.2 Pivoting to Wireshark: Source, Destination, and DNS Update Analysis

The screenshot shows the Wireshark interface with a packet capture of a DNS dynamic update query and response. The packet list shows two packets: a dynamic update query (161 bytes) and a dynamic update response (161 bytes). The packet details pane shows the 'Domain Name System (query)' section, which includes the transaction ID (0xa643), flags (0x2800), and zones (1). The packet bytes pane shows the raw data of the DNS message, including the query and response sections.

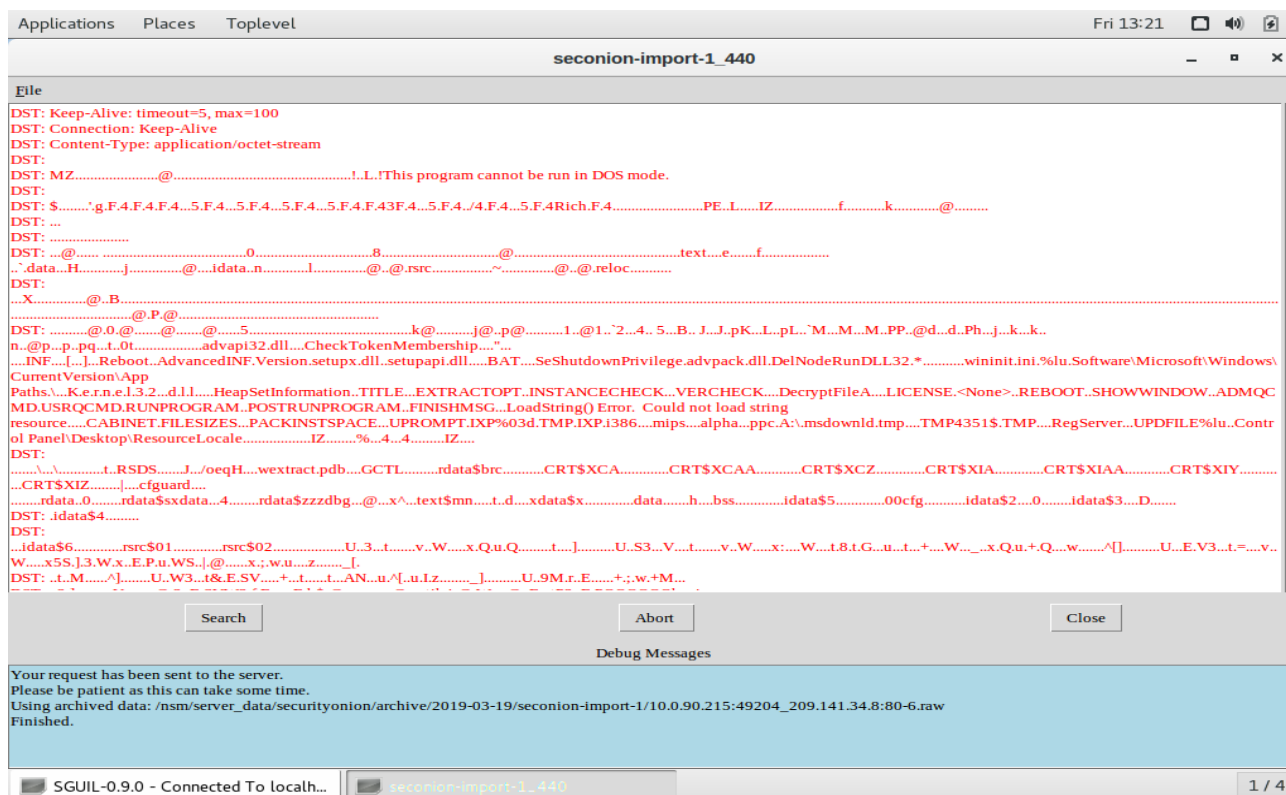
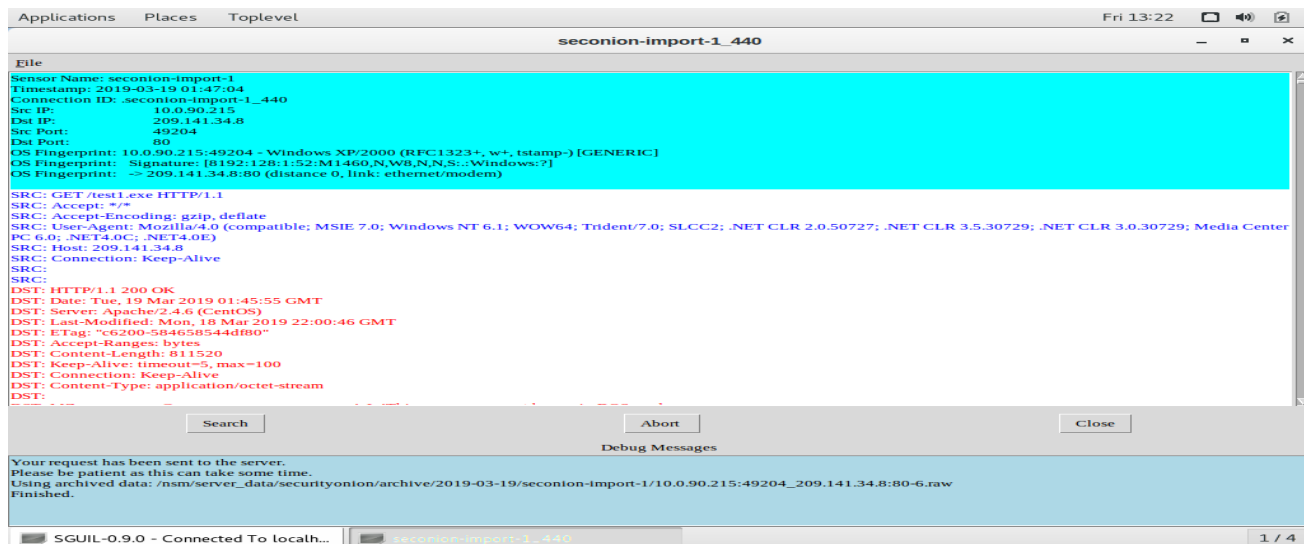
No.	Time	Source	Destination	Protocol	Length	Info
1	2019-03-19 01:45:03.687090	10.0.90.215	10.0.90.9	DNS	161	Dynamic update 0xa643 SOA littletigers.info CNAME AAA
2	2019-03-19 01:45:03.687704	10.0.90.9	10.0.90.215	DNS	161	Dynamic update response 0xa643 SOA littletigers.info ...

Pivoting to Wireshark and filtering on the conversation between 10.0.90.215 and 10.0.90.9 revealed two DNS packets: a dynamic update query and its response. Expanding the Domain Name System layer showed the

update was registering the hostname Bobby-Tiger-PC under the domain littletigers.info. This confirms the DNS update alert was tied to a real internal host attempting to register itself under a suspicious, attacker-controlled domain rather than a normal internal DNS service, suggesting this was a legitimate indicator of compromise rather than an IDS misconfiguration.

1.3 Alert 5.440: Executable Download Confirmed via Transcript Analysis

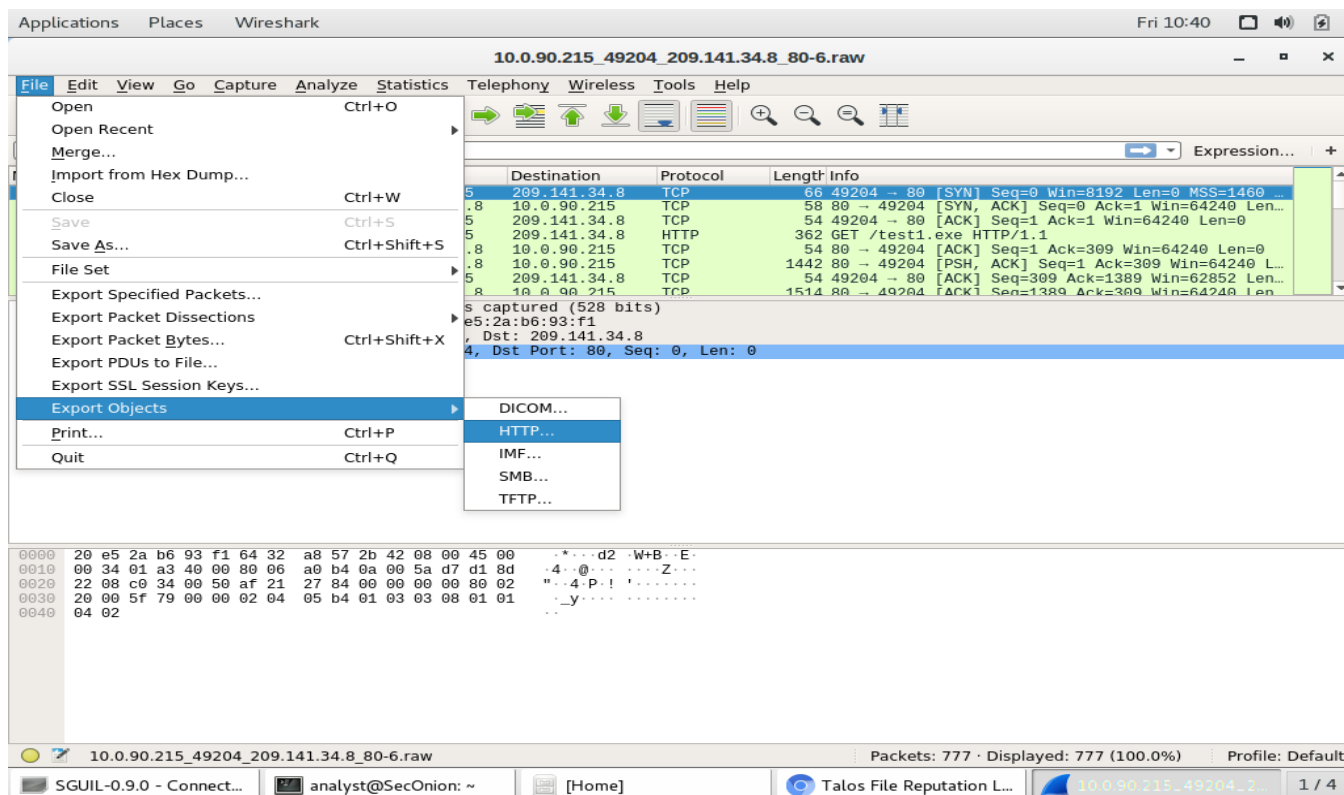
Following up on the suspicious DNS registration from Alert 5.439, the very next alert in the sequence, 5.440, flagged an executable download from the same source IP just two minutes later, making it the natural next step to investigate.



Checking the transcript, I found the host at 10.0.90.215 had sent a GET request for a file called test1.exe over HTTP to 209.141.34.8 on port 80, and the server responded with a 200 OK, delivering a file of 811,520 bytes. Looking at the start of the file's content, I spotted the characters 'MZ', the standard file signature for a Windows executable. This confirmed the host wasn't just behaving suspiciously at the DNS level, it had actually downloaded and likely executed a malicious binary, marking the payload delivery stage of the infection.

1.4 Exporting the Executable from Wireshark for Malware Analysis

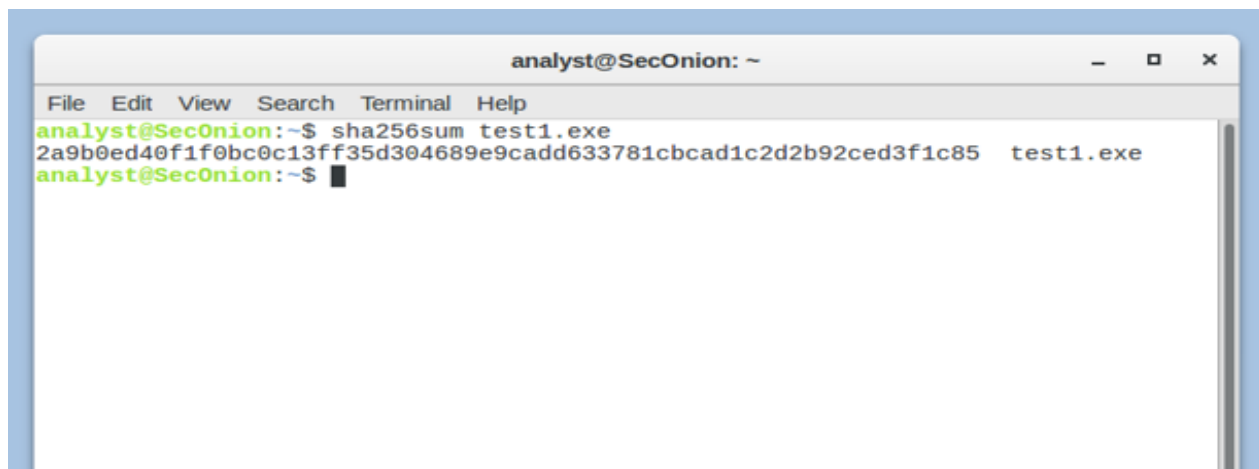
A Wireshark session for Alert 5.440 was opened. I used File > Export Objects > HTTP to extract the downloaded file from the captured traffic and saved it directly to the analyst's home folder for further analysis.



This pulled the executable straight out of the reconstructed TCP stream without needing to touch the raw hex data, saving it as test1.exe in /home/analyst ready for hashing in the next step.

1.5 Generating a SHA256 Hash of the Exported File

With test1.exe saved to the home folder, I opened a terminal and ran sha256sum against the file to generate a unique cryptographic fingerprint that could be used to identify it against known malware databases.



Running sha256sum test1.exe returned the hash: 2a9b0ed40f1f0bc0c13ff35d304689e9cadd633781cbcad1c2d2b92ced3f1c85. This hash acts as a unique identifier for the file, if even a single byte of test1.exe changed, the hash would come out completely different, so it's a reliable way to confirm we're checking the same file that was pulled from the traffic capture.

1.6 Submitting the File Hash to Cisco Talos for Threat Intelligence Lookup

Taking the SHA256 hash generated in the previous step, I submitted it to the Cisco Talos File Reputation Centre at talosintelligence.com to check whether this file had already been identified and catalogued as a known threat.

The screenshot shows the Talos File Reputation Lookup interface. On the left, under 'FILE REPUTATION', there is a biohazard icon and the word 'Malicious'. Below this, the 'TALOS WEIGHTED FILE REPUTATION SCORE' is listed as 'Score not available.' with a 'Submit a File Reputation Ticket' button. On the right, the 'SHA256' hash is displayed: 2A9B0ED40F1F0BC0C13FF35D304689E9CADD633781CBCAD1C2D2B92CED3F1C85. A note states: 'Clicking the above SHA256 will redirect you to Cisco ThreatGrid. This service requires a ThreatGrid subscription.' Below the hash, a table provides file details:

FILE SIZE	811520 bytes
SAMPLE TYPE	PE32 executable (GUI) Intel 80386, for MS Windows
CISCO SECURE ENDPOINT DETECTION NAME	Remcos::gravity::Win.Dropper.Vbkryjetor::in03.talos

Below the table, it says '*Limited to SHA256 lookup'. Further down, 'ASSOCIATED DOMAINS FOR THIS HASH' are listed as 'Domains not available.' and 'DETECTION ALIASES' include '[TROJAN] Trojan/Win32.Generic.C3113109 detected'.

Talos returned a Malicious verdict for this hash. The file was identified as an 811,520-byte PE32 Windows executable, with Cisco Secure Endpoint flagging it under the detection name Remcos::gravity::Win.Dropper.Vbkryjetor::in03.talos. This points to Remcos, a Remote Access Trojan (RAT) commonly delivered through dropper malware, meaning the executable isn't just malicious but is designed to give an attacker remote control over the infected host once run.

This screenshot shows the 'DETECTION ALIASES' section of the Talos File Reputation Lookup page. It lists various identifiers for the file, including '[TROJAN] Trojan/Win32.Generic.C3113109 detected', 'Win32:Evo-gen [Trj]', 'Trojan.GenericKD.42036511', 'win/malicious_confidence_100', 'malware.confidence_100', 'Trojan.Siggen8.18466', 'Generic.mg.f0882b09aebf9b37', 'W32/VBKryjetor.BPIX!tr', 'Trojan.Win32.Krypt', 'Trojan (0054b4501)', 'Generic.Malware/Suspicious', 'RDN/Generic.grp (trojan)', 'BehavesLike.Win32.Generic.bt', 'Trojan:Win32/Wacatac.Hlml', 'Trj/CIA', 'Malware.SwollenFile!1.E38A', and 'MalAgent.J_63873 (Trojan)'.

A quick look through the detection aliases list backs this up, multiple vendors independently flag it as a trojan/dropper (Trojan/Win32.Generic, Win32:Evo-gen [Trj], Trojan.Siggen8), which adds confidence that this isn't a false positive but a genuinely malicious sample tied to the Remcos RAT family.

1.7 Correlating the Talos Detection with the IDS Alert for Remcos RAT Checkin

With Talos confirming test1.exe as Remcos RAT, I checked Sguil for the alert showing it actually communicating. Alert 5.480, "ET TROJAN Remcos RAT Checkin 23," was the one, it names Remcos directly and its count of 404 is far higher than any other alert, pointing to repeated check-ins.

Applications Places Dialog Fri 14:07

SGUIL-0.9.0 - Connected To localhost

File Query Reports Sound: Off ServerName: localhost Username: analyst UserID: 2 2026-07-03 14:07:13 GMT

RealTime Events Escalated Events

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	12	seconion-i...	5.456	2019-03-19 01:47:04	209.141.34.8	80	10.0.90.215	49204	6	ET INFO SUSPICIOUS Dotted Quad Ho...
RT	12	seconion-i...	5.468	2019-03-19 01:47:04	209.141.34.8	80	10.0.90.215	49204	6	ET POLICY PE EXE or DLL Windows fl...
RT	404	seconion-i...	5.480	2019-03-19 01:49:45	10.0.90.215	49205	103.1.184.108	2404	6	ET TROJAN Remcos RAT Checkin 23
RT	1	seconion-i...	5.482	2019-03-19 01:49:46	10.0.90.215	49206	217.23.14.81	80	6	ET CURRENT_EVENTS Terse alphanu...
RT	2	seconion-i...	5.483	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET CURRENT_EVENTS Likely Evil E...
RT	12	seconion-i...	5.485	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET INFO EXE - Served Attached HTTP
RT	12	seconion-i...	5.497	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET INFO Packed Executable Download
RT	12	seconion-i...	5.509	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET CURRENT_EVENTS DRIVEBY Lik...
RT	12	seconion-i...	5.521	2019-03-19 01:49:46	217.23.14.81	80	10.0.90.215	49206	6	ET CURRENT_EVENTS Likely Evil E...

IP Resolution Agent Status Snort Statistics Sy

☐ Reverse DNS ☒ Enable External DNS

Src IP:
 Src Name:
 Dst IP:
 Dst Name:
 Whois Query: ☒ None ☐ Src IP ☐ Dst IP

☒ Show Packet Data ☒ Show Rule

alert tcp \$HOME_NET any -> \$EXTERNAL_NET any (msg:"ET TROJAN Remcos RAT Checkin 23"; flow:established,to_server; dsizel:500; content:"1b 84 d5 b0 5d f4 c4 93 c5 30 c2"; depth:11; fast_pattern; content:"da b1"; distance:2; within:2; threshold:type limit, seconds 30, count 1, track_by_src; metadata: former_category TROJAN; reference:md5,442425e9735892cc9f75711aa8cb210; classtype:trojan-activity; sid:2025637; rev:2; metadata:affected_product Windows_XP_Vista_7_8_10_Server_32_64_Bit,attack_target Client_Endpoint,deployment Perimeter,signature_severity Major,created_at 2018_07_03,updated_at 2018_07_03;) /nsm/server_data/securityion/rules/seconion-import-1/downloaded.rules: Line 20343

IP	Source IP	Dest IP	Ver	HL	TOS	len	ID	Flags	Offset	TTL	ChkSum		
TCP	Source Port	Dest Port	R R R R	1 0	G K H T	N N	Seq #	Ack #	Offset	Res	Window	Urp	ChkSum
	49205	2404	.	.	.	.	346374060	1150900601	5	0	64213	0	55768
DATA	1B 84 D5 B0 5D F4 C4 93 C5 30 C2 C6 8D DA B1 D0 AC AF 6E 7F F8 10 18 23 33 8E D8 54 53 91 AA 53 DB FF 93 6F 0D 73 AF 72 36 40 AD 18 72 20 00 AD 77 D2 1D 5C 2E 33 69 52 12 16 75 A6 DD A9 8E 58 #3 S												

Search Packet Payload ☐ Hex ☒ Text ☐ NoCase

SGUIL-0.9.0 - Connect [analyst@SecOnion: ~] [Home] [Talos File Reputation ...] [10.0.90.215_49204_...] 1 / 4

Looking at the Snort rule, it isn't triggering on the source IP or port, it's matching a specific hex byte pattern (1b 84 d5 b0 5d f4 c4 93 c5 30 c2) found right at the start of the TCP payload, visible in the hex pane. This pattern is like a known fingerprint Remcos leaves behind when it checks in with its command-and-control (C2) server, so seeing it here confirms the malware was actually running and talking back to the attacker.

1.8 Transcript Analysis: Confirming Active Remcos RAT Command and Control Communication

With Talos confirming the file as Remcos RAT and Sguil's Alert 5.480 showing it actively checking in, the next logical step was to open the transcript for that alert and examine what the actual C2 communication looked like on the wire.

Applications Places Toplevel Fri 18:45

seconion-import-1_480

File

Sensor Name: seconion-import-1
Timestamp: 2019-03-19 01:49:45
Connection ID: seconion-import-1_480
Src IP: 10.0.90.215
Dst IP: 103.1.184.108
Src Port: 49205
Dst Port: 2404
OS Fingerprint: 10.0.90.215:49205 - Windows XP/2000 (RFC1323+, w+, tstamp-) [GENERIC]
OS Fingerprint: Signature: [8192:128:1:52:M1460,NW8,N,N,S:;Windows:?] OS Fingerprint: -> 103.1.184.108:2404 (distance 0, link: ethernet/modem)

```

SRC: [...]0.....0...3*...k...
SRC: b...T...z...OT...z%K...&?%...*...9.....4.F...T...o...3IBQ[...C[...p...j.../...T>d...>..
SRC:
B.....IA..M{...JMc...#...o...n>...y...$@{(0...&...o...6...{c...RLe..._...e.....l...[4]6[gn...y?;...^...C.B'.....0...C...4.G.m3...l{C.L.Z...=...z...y.N...T...o...NlZr...0...Co...X...p.s.hk...J...+...$...2...;...f...v}f.r
a.00.D^...(.lp...B...M...M...^...f...d...)*...Ck
SRC: [...]31.e...3...S0.....2e.....lq.9...:({
SRC: [...]0...O...<...PxS...B...R...l...L.96...k...m...f...a.a.4.RW.k...M...0)...=(2...88...f...a...m...Y...m=Q...uy.TL.qy...E'([...b].....j...m"G...
DST: [...]0.....0.....#3...
SRC: [...]0.....0.....#3...TS...S...o
SRC: s.r6@r...w...3IR...u...X.H...l...T.qb@'.GO&R...j...PP&H...).sHK...~...w...o...R...l...
DST: [...]0.....0.....#3...
DST: >...7...nqe.v6r.Tp...w.YMp
DST: Zz...*...9...L4.F...l...fDc+z...@...2S..4P...
DST: [...]0.....0.....#3...
SRC: [...]0.....0.....#3...TS...S...o
SRC: s.r6@r...w...3IR...u...X.H...l...T.qb@'.GO&R...j...PP&H...).sHK...~...w...o...R...l...
DST: [...]0.....0.....#3...
SRC: [...]0.....0.....#3...TS...S...o
SRC: s.r6@r...w...3IR...u...X.H...l...T.qb@'.GO&R...j...PP&H...).sHK...a7N...1..V.C:
DST: [...]0.....0.....#3...

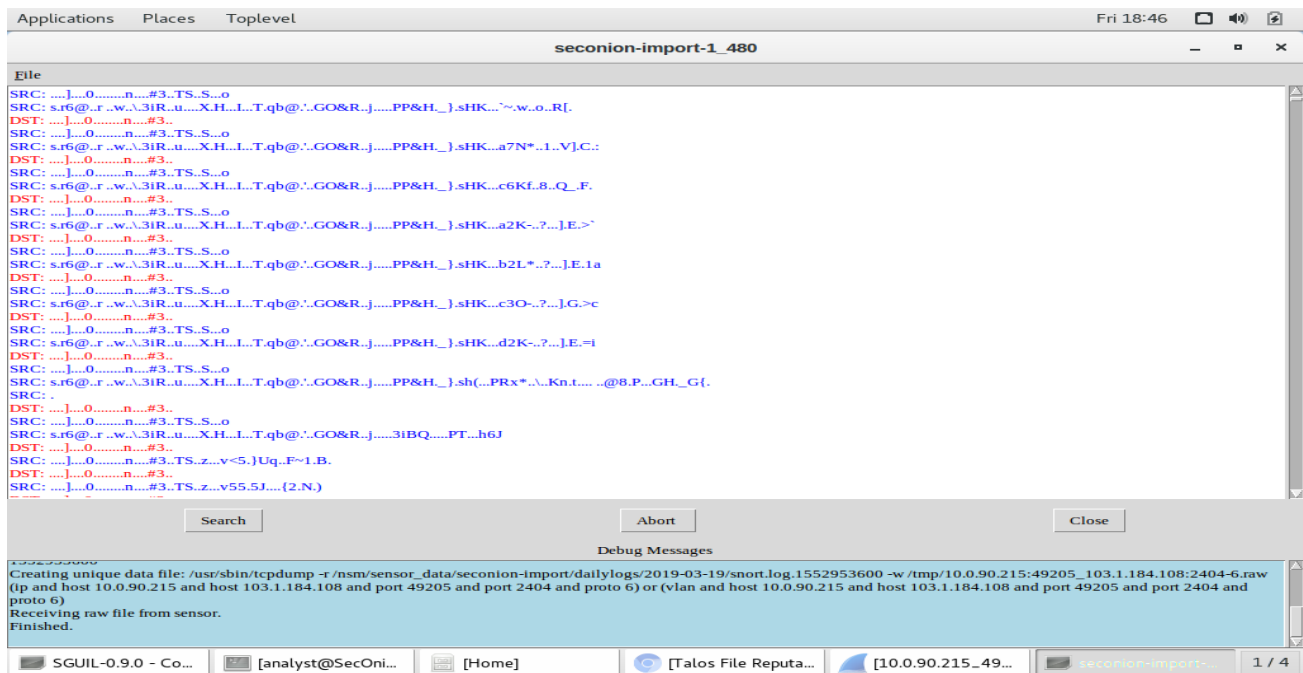
```

Search Abort Close

Debug Messages

Creating unique data file: /usr/sbin/tcpdump -r /nsm/sensor_data/seconion-import/dailylogs/2019-03-19/snort.log.1552953600 -w /tmp/10.0.90.215:49205_103.1.184.108:2404-6.raw (ip and host 10.0.90.215 and host 103.1.184.108 and port 49205 and port 2404 and proto 6) or (vlan and host 10.0.90.215 and host 103.1.184.108 and port 49205 and port 2404 and proto 6)
Receiving raw file from sensor.
Finished.

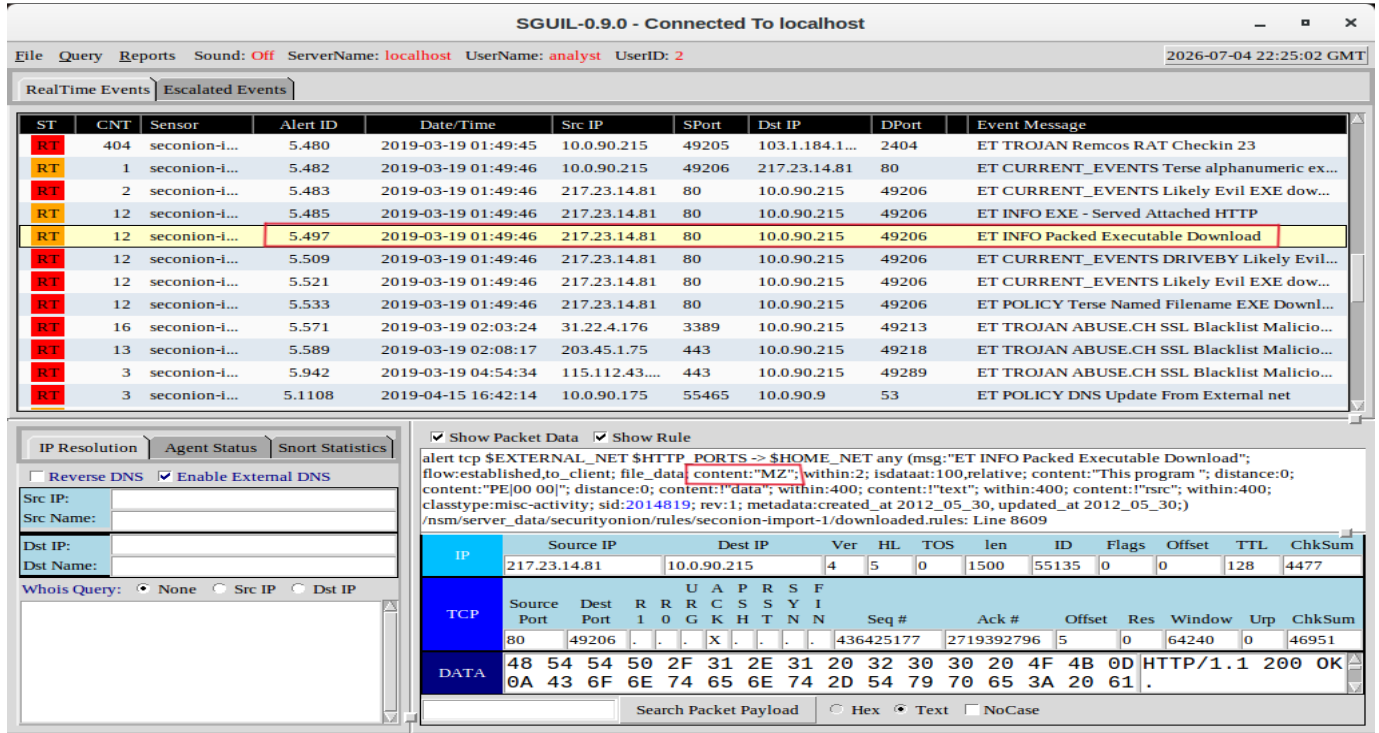
SGUIL-0.9.0 - Co... [analyst@SecOni... [Home] [Talos File Reputa... [10.0.90.215_49... seconion-import-1_480 1 / 4



The connection ran from 10.0.90.215 (port 49205) to 103.1.184.108 on port 2404, an obscure, non-standard port that legitimate software rarely uses for outbound traffic. Scrolling through the transcript, the content exchanged between the two hosts was heavily obfuscated and completely unreadable, confirming the C2 traffic was encrypted to avoid detection. Remcos, which stands for Remote Control and Surveillance, uses custom encoding on its check-in traffic specifically to blend in and bypass signature-based tools. This confirmed the infected host wasn't just carrying the malware, it was actively running it and maintaining a live communication channel back to the attacker's server.

1.9 Alert 5.497: Investigating a Second Packed Executable Download

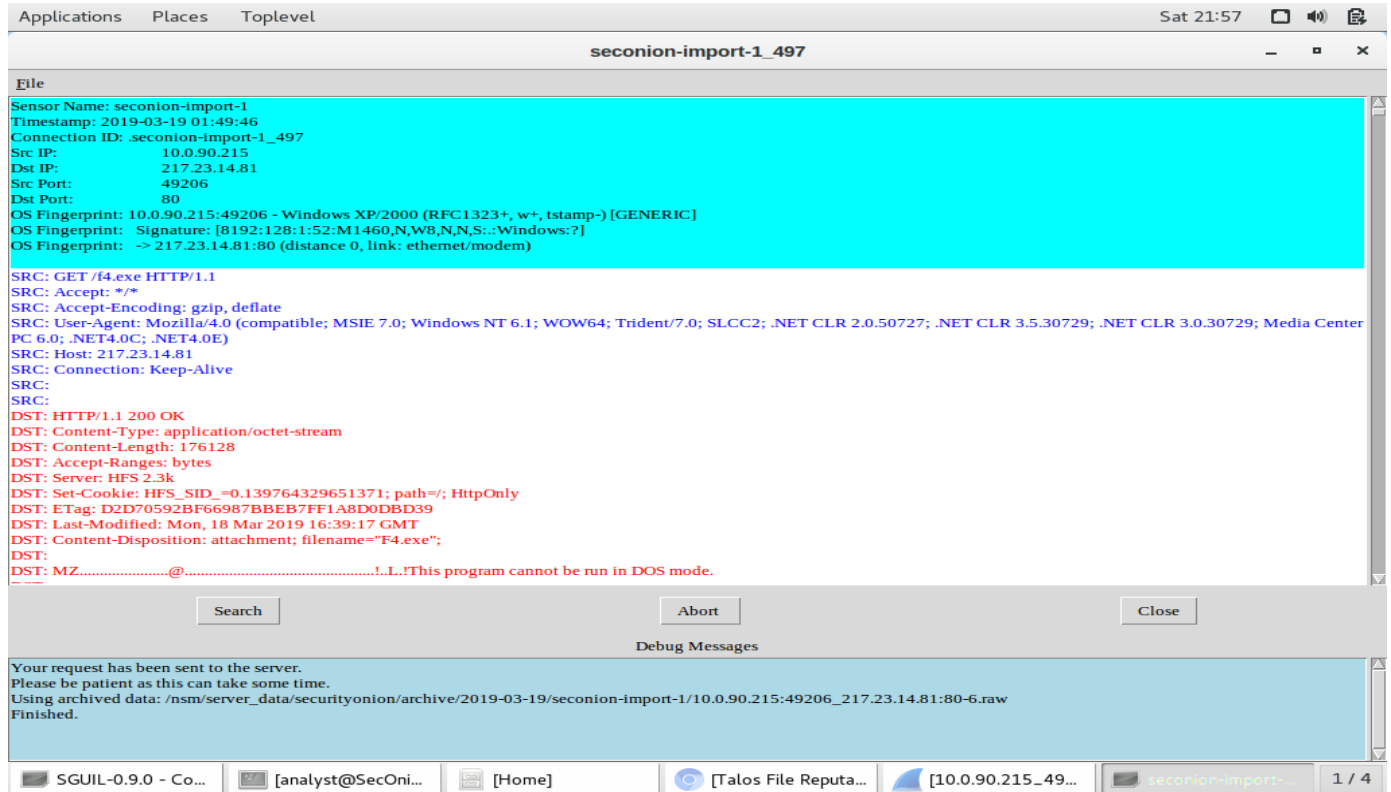
Reviewing the remaining alerts from 19 March 2019, Alert ID 5.497 stood out with an event message reading 'ET INFO Packed Executable Download', indicating another executable file was pulled down to the infected host. I selected it for further investigation to determine whether this was a second payload or part of the same infection chain.



Selecting Alert ID 5.497 and reviewing the packet data and Snort rule revealed the signature 'ET INFO Packed Executable Download' (SID 2014819), triggered by two content matches: the 'MZ' Windows executable file signature and the string 'This program', both standard markers of a Windows PE file being transferred over HTTP. The source IP 217.23.14.81 on port 80 delivered the file to the infected host at 10.0.90.215 on port 49206, with the HTTP 200 OK response visible in the data layer confirming the transfer was already complete.

1.10 Transcript Analysis: Confirming the Second Executable Download (f4.exe)

I opened the transcript for a closer look at the actual request and server response.



Opening the transcript for Alert 5.497 confirmed the infected host at 10.0.90.215 sent a GET request for a file named f4.exe from 217.23.14.81 on port 80. The server responded with a 200 OK, delivering a 176KB file with Content-Type: application/octet-stream. The filename f4.exe in the Content-Disposition header and the MZ signature visible at the bottom of the response leave no doubt, this was a second Windows executable delivered to the already compromised host, deepening the infection.

1.11 Exporting and Hashing f4.exe for Threat Intelligence Lookup

Following the same process as test1.exe, I exported f4.exe from Wireshark via File > Export Objects > HTTP and saved it to the analyst's home folder.



Running sha256sum against both files in the terminal confirmed their unique cryptographic fingerprints. f4.exe produced the hash 5865e801e6324166d6d05b39a14f2a8a798c6eb652831f78c2634f2b7a400eaf, a completely different signature from test1.exe, confirming these are two distinct malware samples rather than the same file downloaded twice.

1.12 Talos Threat Intelligence Lookup: Identifying f4.exe as a Known Dropper

I submitted the SHA256 hash of f4.exe to Cisco Talos File Reputation Centre to check whether this second executable had already been catalogued as a known threat.

Talos File Reputation Lookup || Cisco Talos Intelligence Group - Comprehensive Threat Intelligence - Chromium

Talos File Reputation Lookup x +

talosintelligence.com/sha_searches

This system limits you to one lookup at a time, and is limited to only hash matching.

FILE REPUTATION



Malicious

SHA256

5865E801E6324166D6D05B39A14F2A8A798C6EB652831F78C2634F2B7A400EAF

Clicking the above SHA256 will redirect you to Cisco ThreatGrid. This service requires a ThreatGrid subscription.

FILE SIZE	176128 bytes
SAMPLE TYPE	PE32 executable (console) Intel 80386, for MS Windows
CISCO SECURE ENDPOINT DETECTION NAME	Win.Dropper.Cridex::1201

TALOS WEIGHTED FILE REPUTATION SCORE ⓘ

Score not available.

ASSOCIATED DOMAINS FOR THIS HASH

Domains not available.

*Limited to SHA256 lookup

Talos returned a clear Malicious verdict. The file is a 176,128-byte PE32 Windows executable, and the Cisco Secure Endpoint detection name is Win.Dropper.Cridex::1201, identifying it as a Cridex dropper. Cridex is a well-known banking trojan that steals financial credentials and can download additional malware onto the infected host. Combined with the Remcos RAT already confirmed on this machine, this points to a deliberate two-stage attack: Remcos provides the attacker with remote access and control, while Cridex handles credential theft and financial fraud.

1.13 SSL Blacklist Alerts: Identifying Redundant C2 Channels via Malicious Certificate Detection

Shifting focus to the three SSL blacklist alerts from 19 March 2019, all sharing the same event message: ET TROJAN ABUSE.CH SSL Blacklist Malicious SSL Certificate Detected, I opened each transcript to investigate how they were connected.

Applications Places Toplevel Sun 04:18

seconion-import-1_571

File

Sensor Name: seconion-import-1
Timestamp: 2019-03-19 02:03:24
Connection ID: .seconion-import-1_571
Src IP: 10.0.90.215
Dst IP: 31.22.4.176
Src Port: 49213
Dst Port: 3389
OS Fingerprint: 10.0.90.215:49213 - Windows XP/2000 (RFC1323+, w+, tstamp-) [GENERIC]
OS Fingerprint: Signature: [8192:128:1:52:M1460,N,W8,N,N,S::Windows:?]
OS Fingerprint: -> 31.22.4.176:3389 (distance 0, link: ethernet/modem)

SRC:[.\\M.f].....#M...0:.....X...*,<./,=5...
SRC: '.....+.#,,\$...
SRC: .@.2.j.8.....)
SRC:
SRC:
DST: ...Q...M...\\M.a.(.GaQ...9J...W NO.4:16:....w...-Xld#T.....~./0...0.m.....0
DST: ..*..H..
DST:0Y1.0...U...MV1
DST: 0...U...Male1.0...U..
DST: ..Wsiaen LLC.1%0#..U...3Arefotoutoora.2Lecenwin.mz0..
DST: 190318075829Z..
DST: 190916075829Z0Y1.0...U...MV1
DST: 0...U...Male1.0...U..
DST: ..Wsiaen LLC.1%0#..U...3Arefotoutoora.2Lecenwin.mz0.."0
DST: ..*..H..
DST:0..
DST:
DST: ...[.s[.6]eR.....Q.2?..8....a0..K...7~"@x.#...M.d..F8.L...x0LP...Zxc.....R.(.(\\C...V{..1...Q..x..\$^...6....^P...`.....j.S...[.].`-7...[Y..n!.....W*..-@.....
DST: .[5.m.-.....*..T.dX...[.E.....P0N0...U.....A...@sB..9cD..os.i0...U...#..0...-A..@sB..9cD..os.i0...U...0...0
DST: ..*..H..

Search Abort Close

Debug Messages

Your request has been sent to the server.
Please be patient as this can take some time.
Raw data request sent to seconion-import-1.
Making a list of local log files.
Looking in /nsm/sensor_data/seconion-import/dailylogs/2019-03-19.
Making a list of local log files in /nsm/sensor_data/seconion-import/dailylogs/2019-03-19.

[SGUIL-0... [analyst... [Home] [Talos Fi... [10.0.90... seconion... seconion... seconion... seconion... 1 / 4

Alert 5.571

Applications Places Toplevel Sun 04:19

seconion-import-1_942

File

Sensor Name: seconion-import-1
Timestamp: 2019-03-19 04:54:34
Connection ID: .seconion-import-1_942
Src IP: 10.0.90.215
Dst IP: 115.112.43.81
Src Port: 49289
Dst Port: 443
OS Fingerprint: 10.0.90.215:49289 - Windows XP/2000 (RFC1323+, w+, tstamp-) [GENERIC]
OS Fingerprint: Signature: [8192:128:1:52:M1460,N,W8,N,N,S::Windows:?]
OS Fingerprint: -> 115.112.43.81:443 (distance 0, link: ethernet/modem)

SRC:[.\\V...]=Sd%....9...1..qz*@V.....*,<./,=5...
SRC: '.....+.#,,\$...
SRC: .@.2.j.8.....)
SRC:
SRC:
DST: ...Q...M...\\u..i..B..&8...9*..8w.G.,>.. ..\\.....1...LZ...s..k..d./.....%...!.....0...0.....E..0
DST: ..*..H..
DST:0..1.0...U...IL1\$0"..U...Witba oredorya8 dtjen siral1.0...U...Tel Aviv1.0...U..
DST: ..Pratherl NL1.0...U...Ganof and Vititiraz1%0#..U...Vedabenm.cabermasthttan.sarl0..
DST: 190311192959Z..
DST: 190909192959Z0..1.0...U...IL1\$0"..U...Witba oredorya8 dtjen siral1.0...U...Tel Aviv1.0...U..
DST: ..Pratherl NL1.0...U...Ganof and Vititiraz1%0#..U...Vedabenm.cabermasthttan.sarl0.."0
DST: ..*..H..
DST:0..
DST:
DST:r]...7mF...n.Y...C...cb...W.K..'k%6u...+..cg6.O.f...Aw...E...F[e{.z...t.6.j)}*mW...U...EU.../..X.x9...u.....u...{..#Fn>...N..t.s.f).r.{C...[...H.F[...O...[...B...O"...a.O.6.K...
DST: L.q.
DST: ...nH..Y[Z.V1/c.[.....G...b5.Z...K...=.....P0N0...U.....t).9..f.=nz.z0...U...#..0.....t).9..f.=nz.z0...U...0...0
DST: ..*..H..
DST:Ea.7J..6...C...=>...[.....F.y^(...D.p...c..T|<la%J.1...I...0...[...tD..L.LL.....g]'_ai3..iH....{..z,

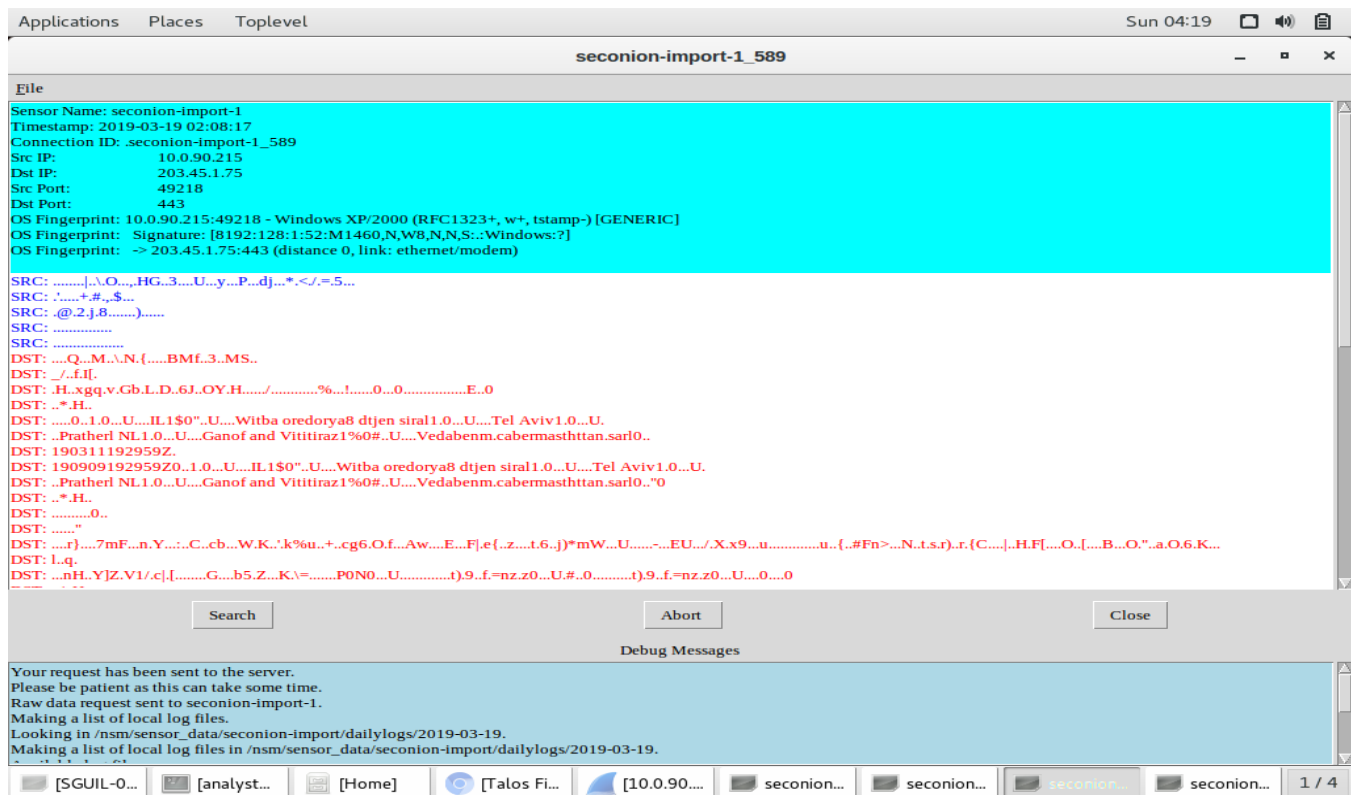
Search Abort Close

Debug Messages

1552953600
Creating unique data file: /usr/sbin/tcpdump -r /nsm/sensor_data/seconion-import/dailylogs/2019-03-19/snort.log.1552953600 -w /tmp/10.0.90.215:49289_115.112.43.81:443-6.raw (ip and host 115.112.43.81 and host 10.0.90.215 and port 443 and port 49289 and proto 6) or (vlan and host 115.112.43.81 and host 10.0.90.215 and port 443 and port 49289 and proto 6)
Receiving raw file from sensor.
Finished.

[SGUIL-0... [analyst... [Home] [Talos Fi... [10.0.90... seconion... seconion... seconion... seconion... 1 / 4

Alert 5.942



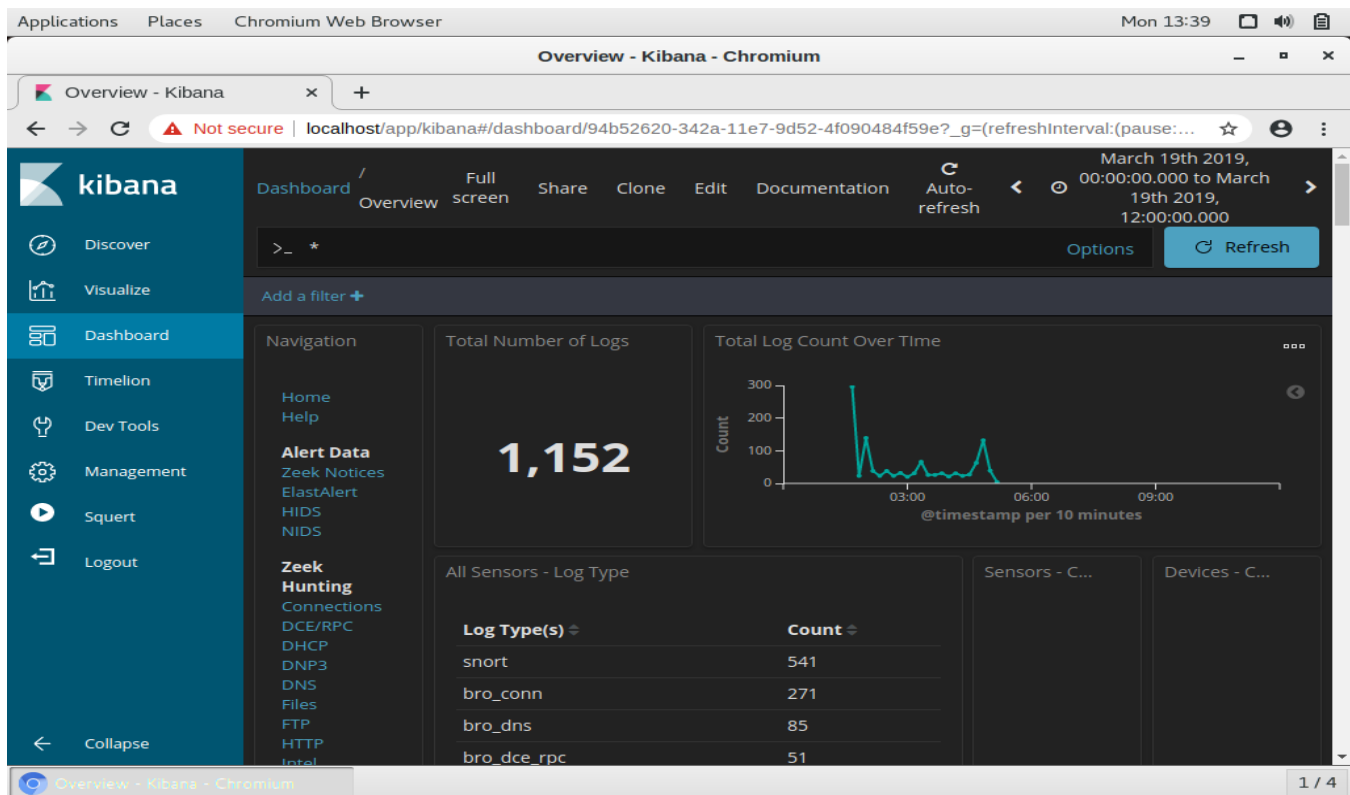
Alert 5.589

Looking at all three transcripts, the infected host at 10.0.90.215 initiated outbound encrypted connections to three different external IPs (31.22.4.176 on port 3389, 203.45.1.75 on port 443, and 115.112.43.81 on port 443) at different points across the attack timeline. Critically, all three were triggered by the same Snort rule (SID 2023476), which targets malicious SSL certificates associated with Dridex, confirming this wasn't three separate threats but one malware maintaining multiple redundant C2 channels to different servers. This maps to the Command-and-Control phase of the Cyber Kill Chain, where the attacker ensures persistence by distributing outbound communication across multiple endpoints, so that blocking one doesn't cut off access entirely.

Part 2: Correlating the Attack Timeline in Kibana

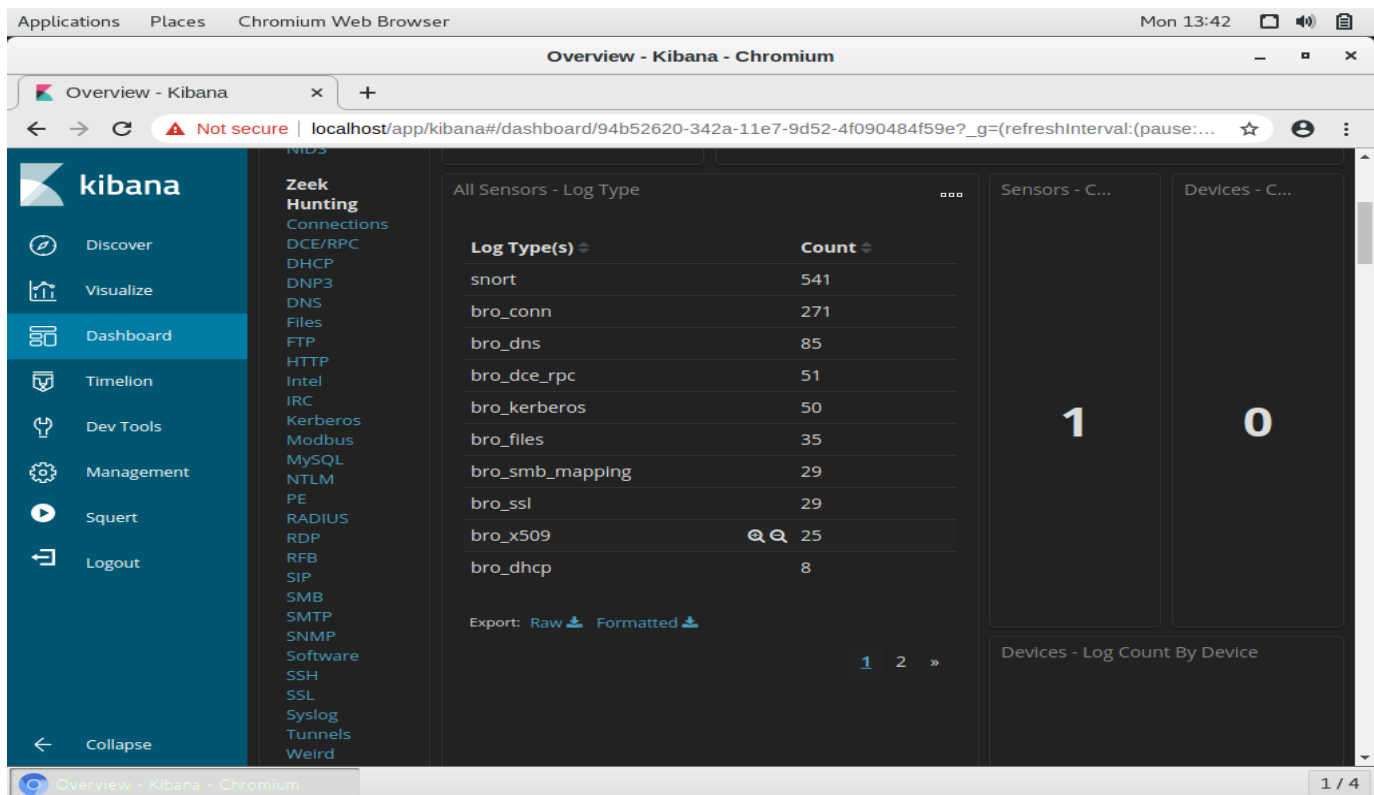
2.1 Correlating the Attack Timeline in Kibana

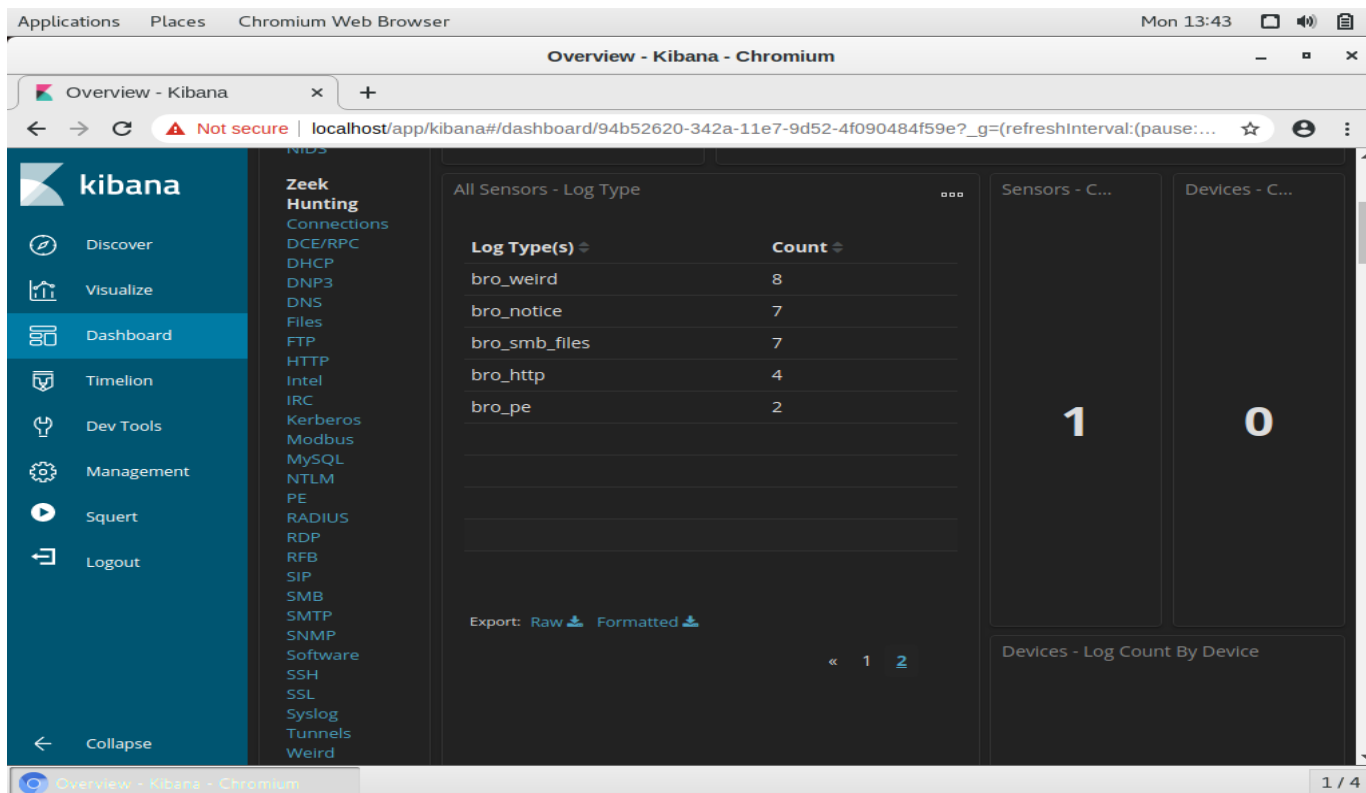
Every alert investigated so far, from the initial DNS update through the executable downloads, Remcos RAT check-ins, and SSL blacklist detections, traced back to the same infected host: 10.0.90.215, confirmed as Bobby-Tiger-PC through the DNS registration in Alert 5.439. What remained unconfirmed was the actual Windows username logged into that machine. I moved into Kibana, set the time range to March 2019, and the log count timeline spiked on 19 March, narrowing the focus to the exact attack window to pin down that remaining detail.



2.2 Kibana Log Type Review: Mapping the Full Scope of Activity Beyond Sguil's Alerts

With the timeframe narrowed to the attack window, the next step was to check the All Sensors, Log Type panel to see every log source tied to this single infection on Bobby-Tiger-PC, not just the alerts already reviewed in Sguil, since alerts only capture activity that triggered a rule, while other log types can reveal related traffic that didn't.

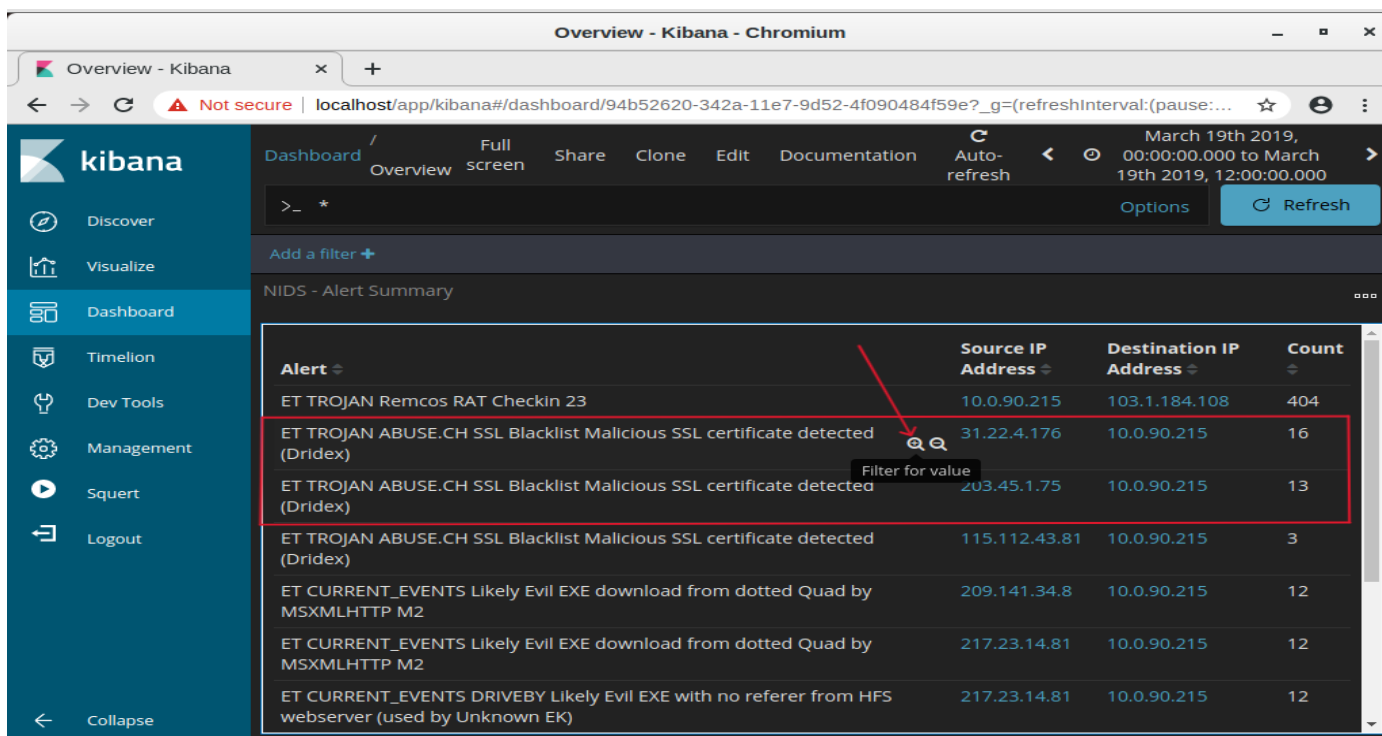




Across both pages, the log types ranged from snort (541) and bro_conn (271) down to bro_dns, bro_files, bro_ssl, and bro_x509 on the first page, and bro_weird, bro_notice, bro_smb_files, bro_http, and bro_pe on the second. This confirmed the infection touched far more of the network stack than the handful of alerts alone suggested, giving a fuller picture of the host's activity during the attack window.

2.3 Filtering the NIDS Alert Summary for SSL Blacklist Activity

After reviewing the full range of log types available for this incident, the next step was to narrow back in on the SSL blacklist alerts specifically. In the NIDS Alert Summary table, I clicked the magnifying glass to filter the entire dashboard down to just the 'ET TROJAN ABUSE.CH SSL Blacklist Malicious SSL certificate detected (Dridex)' alert, since this was the one area from Sguil we hadn't dug deeper into yet.



Filtering by this alert updated the dashboard to show only the three SSL blacklist entries, 32 logs in total across the three attacker IPs (31.22.4.176, 203.45.1.75, and 115.112.43.81). This narrows the view from just knowing an alert fired to seeing every connection behind it, setting up the next step of digging into attacker location. It fits the Detection and Analysis phase of incident response, and ties to the Command and Control stage of the Cyber Kill Chain.

2.4 Expanding Log Entries: Geolocating the Three Dridex C2 Servers

With the dashboard now filtered to just the 32 SSL blacklist logs, the next step was to expand each entry by clicking the arrow beside it to reveal the full log details, including geolocation data for each attacker IP.

Overview - Kibana - Chromium

Overview - Kibana

Dashboard / Overview Full screen Share Clone Edit Documentation Auto-refresh

March 19th 2019, 00:00:00.000 to March 19th 2019, 12:00:00.000 Options Refresh

alert.keyword: "ET TROJAN ABUSE.CH SSL Blacklist Malicious SSL certificate detected (Dridex)" Add a filter Actions

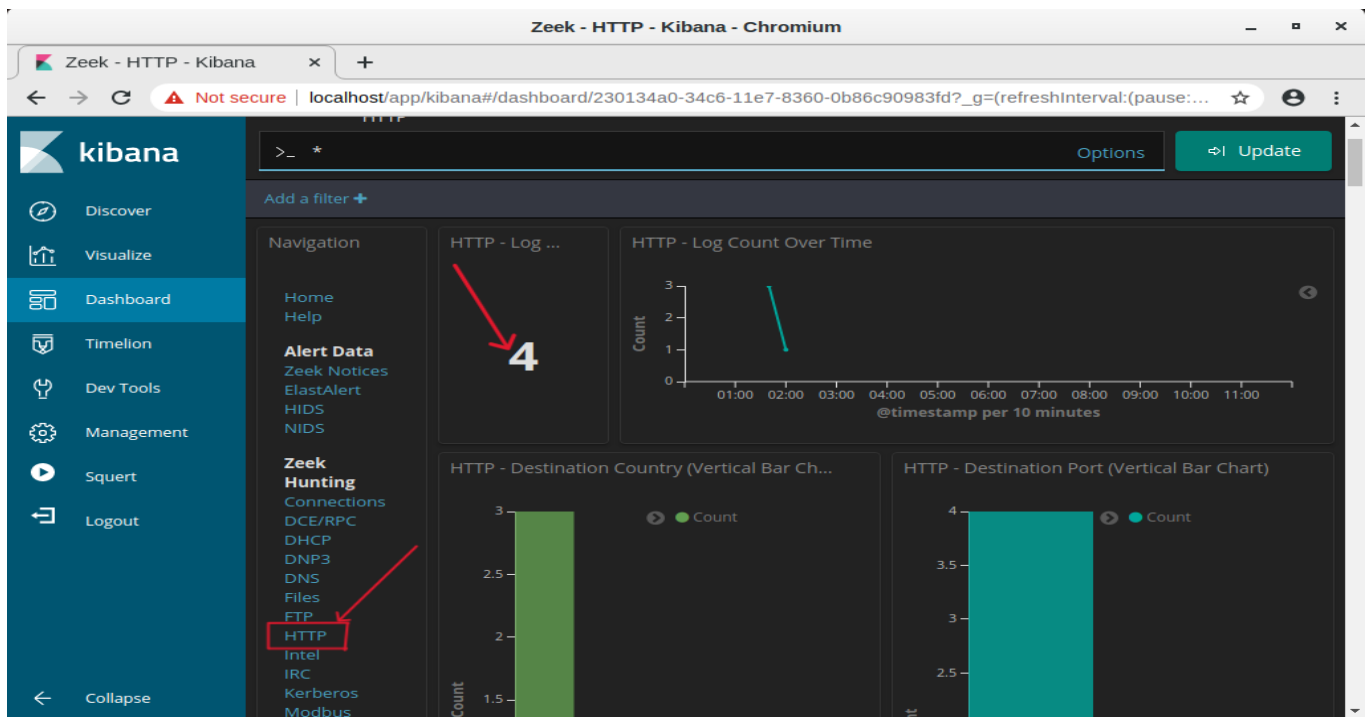
All Logs Limited to 10 results. Refine your search. 1-10 of 32

Time	source_ip	source_port	destination_ip	destination_port	_id
▶ March 19th 2019, 04:55:13.000	115.112.43.81	443	10.0.90.215	49298	1OGzA3MB7nV0xlRopFlx
▶ March 19th 2019, 04:54:57.000	115.112.43.81	443	10.0.90.215	49295	0-GzA3MB7nV0xlRopFlx
▶ March 19th 2019, 04:54:34.000	115.112.43.81	443	10.0.90.215	49289	0eGzA3MB7nV0xlRopFlx
▶ March 19th 2019, 04:50:21.000	31.22.4.176	3389	10.0.90.215	49281	xuGzA3MB7nV0xlRoolkK
▶ March 19th 2019, 04:50:21.000	31.22.4.176	3389	10.0.90.215	49281	x-GzA3MB7nV0xlRopFlx

Expanding the logs for all three source IPs revealed their real-world locations: 203.45.1.75 traced to Mount Waverley, Australia, 31.22.4.176 traced to Newcastle upon Tyne, United Kingdom, and 115.112.43.81 traced to Mumbai, India. All three connections shared the same Snort rule (SID 2023476) and severity rating of High, confirming they're part of the same Dridex C2 infrastructure, just distributed across three different countries, likely to make takedown or blocking harder for defenders.

2.5 HTTP Dashboard Review: Separating Malicious Downloads from Legitimate Windows Traffic

With the SSL blacklist connections geolocated, the next step was to check the built-in HTTP dashboard under Zeek Hunting to see everything the infected host requested over plain HTTP, since unencrypted traffic can reveal details SSL traffic hides.



The HTTP dashboard logged 4 requests total, from the United States and Netherlands.

HTTP - URIs

URI	Count
/f4.exe	1
/ncsl.txt	1
/pki/crl/products/CSPCA.crl	1
/test1.exe	1

Count - MIME Type

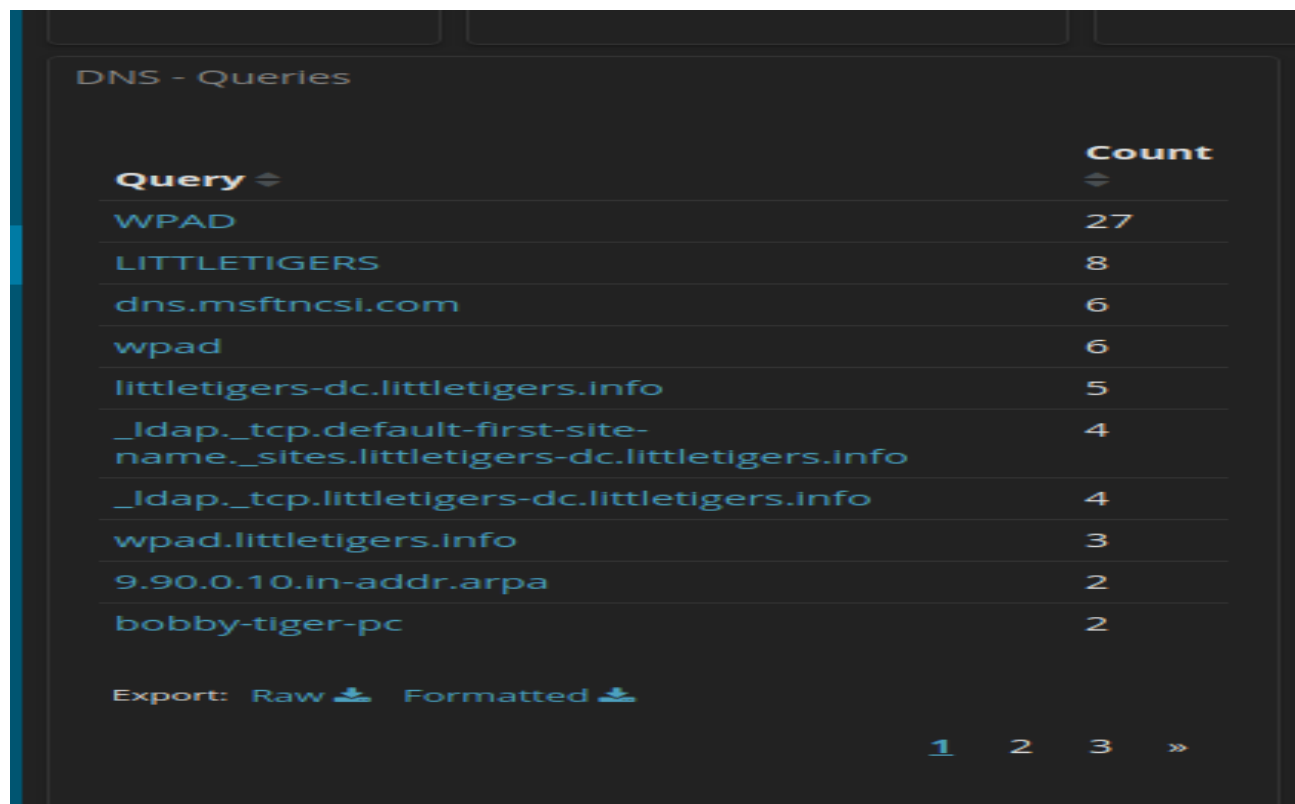
HTTP - Sites

Site	Count
209.141.34.8	1
217.23.14.81	1
crl.microsoft.com	1
www.msftncsl.com	1

To match each URI to its source, I filtered on each entry in the HTTP-URIs panel individually, which narrowed the HTTP-Sites panel down to show exactly which site that specific request came from. This confirmed /test1.exe and /f4.exe were requested from 209.141.34.8 and 217.23.14.81, the two malicious sites already tied to Remcos RAT and Cridex. The remaining two, /ncsi.txt and /pki/crl/products/CSPCA.crl, matched to www.msftncsi.com and crl.microsoft.com, both routine Windows background checks (internet connectivity and certificate revocation), unrelated to the attack. This ruled out any third hidden payload among the HTTP logs.

2.6 DNS Query Review: Identifying a Suspicious New Domain (toptoptop1.online)

Since the investigation started with a single DNS update tied to littletigers.info, the next step was to check the full DNS Queries panel for the complete picture, not just that one alert.



Query	Count
WPAD	27
LITTLETIGERS	8
dns.msftncsi.com	6
wpad	6
littletigers-dc.littletigers.info	5
_ldap._tcp.default-first-site-name._sites.littletigers-dc.littletigers.info	4
_ldap._tcp.littletigers-dc.littletigers.info	4
wpad.littletigers.info	3
9.90.0.10.in-addr.arpa	2
bobby-tiger-pc	2

Export: Raw Formatted

1 2 3 »



Query	Count
isatap.localdomain	1
toptoptop1.online	1
www.msftncsi.com	1

Export: Raw Formatted

« 1 2 3

Page 1 showed WPAD and littletigers.info dominating the query list, consistent with what we already knew. On page 3, two entries were quickly explained, isatap.localdomain (a standard Windows mechanism) and www.msftncsi.com (already confirmed as Microsoft's connectivity check). The third, toptoptop1.online, hadn't appeared anywhere else, making it worth looking up.

The screenshot shows the VirusTotal interface for the URL <http://toptoptop1.online/>. At the top, a red banner indicates that 17 out of 92 security vendors flagged the URL as malicious. A circular gauge on the left shows the 'Community Score' as 17/92. Below this, the 'DETECTION' tab is active, displaying a 'Crowdsourced context' section with a warning icon and text: 'ThreatFox IOCs for 2023-09-10 - according to source ArcSight Threat Intelligence - 2 years ago' and 'Quasar RAT botnet C2 domain (confidence level: 100%)'. The 'Security vendors' analysis' section shows a table of detections from various vendors.

Security vendors' analysis		Do you want to automate checks?	
ADMINUSLabs	Malicious	alphaMountain.ai	Malicious
AlphaSOC	Malware	Antiy-AVL	Malicious
BitDefender	Phishing	Chong Lua Dao	Malicious

Submitting toptoptop1.online to VirusTotal confirmed it as malicious, 17 out of 92 vendors flagged it, with several labelling it malware or phishing. ThreatFox identified it with 100% confidence as a Quasar RAT botnet C2 domain, adding a third distinct malware family to the investigation alongside Remcos RAT and Cridex.

Part 3: Findings, Lessons Learned, and Recommendations

3.1 Mitigation Measures and Post-Incident Security Hardening

To contain this infection, I recommend immediately isolating the host (Bobby-Tiger-PC) from the network to cut off all three active C2 channels. All credentials used on this machine should be reset, since Cridex is designed to harvest banking and login credentials. The malicious domains and IPs identified (littletigers.info, toptoptop1.online, and the three SSL blacklist IPs) should be blocked at the firewall and added to DNS sinkholing rules. Endpoint detection should be deployed to catch the Remcos, Cridex, and Quasar RAT binaries directly, and WPAD should be disabled or tightly restricted unless explicitly required, since it's a common vector for traffic interception. Finally, a full forensic disk image of the host should be taken before any remediation, to preserve evidence and check for additional persistence mechanisms.

3.2 Observations and Reflections

This lab showed how a single infected host can end up compromised by multiple, unrelated malware families at once, not just one attacker, but at least three separate threats (Remcos RAT, Cridex, and Quasar RAT) all operating on the same machine. What stood out most was how much of the attack was hidden in plain sight: DNS queries mimicking legitimate Active Directory traffic, encrypted SSL connections blending in as normal HTTPS, and background Windows processes (NCSI, CRL checks) sitting right alongside genuinely malicious activity. It reinforced that an analyst can't just trust a single alert or a single tool, the full picture only came together by cross-referencing Sguil, Wireshark, and Kibana together.

3.3 What I Learned

I learned how to:

- Trace a multi-stage malware infection from a single DNS alert through to full command-and-control confirmation
- Extract and hash files directly from Wireshark for threat intelligence lookups
- Use Cisco Talos and VirusTotal to identify malware families and C2 infrastructure from file hashes and domains
- Correlate alerts, geolocation data, and log types in Kibana to build a complete attack timeline
- Distinguish real malicious activity from legitimate background Windows traffic

3.4 Skills Demonstrated

- Multi-tool threat correlation across Sguil, Wireshark, and Kibana
- File hash generation and threat intelligence lookup (SHA256, Cisco Talos, VirusTotal)
- DNS and HTTP traffic analysis to separate malicious from benign activity
- Threat attribution and malware family identification (Remcos RAT, Cridex, Quasar RAT)
- Geolocation-based infrastructure analysis of C2 servers
- Structured incident investigation aligned to the Cyber Kill Chain

3.5 Conclusion

This lab demonstrated how a single compromised host can carry multiple distinct malware families simultaneously, each with its own delivery method, communication channel, and objective. Investigating this attack end to end, from an initial DNS registration to confirmed RAT and banking trojan activity, showed how SOC analysts must move fluidly between tools to build a complete picture, since no single log source tells the whole story on its own. These correlation and threat attribution skills translate directly to real-world SOC Tier 1 responsibilities, where quickly distinguishing genuine threats from background noise is critical to effective incident response.